

Pushkar Singhania

Chennai, India

+91 7866910932 — pushkar.singhania3093@gmail.com — GitHub — LinkedIn

Summary

Electrical and Computer Science undergraduate at VIT Chennai (CGPA: 9.07/10) with strong foundations in **Java, Python, Data Structures, Algorithms, Object-Oriented Programming, SQL, and Software Development**. Experienced in designing and delivering **full-stack applications, REST APIs, and machine learning systems**. Passionate about building scalable, resilient technology solutions and collaborating in large, diverse engineering teams to solve real-world problems.

Education

Vellore Institute of Technology, Chennai	2024 – 2028
B.Tech in Electrical and Computer Science Engineering	CGPA: 9.07 / 10
GD Birla Centre for Education, Kolkata	Class XII (2024)
Higher Secondary Education	92%
Sacred Heart School	Class X (2022)
Secondary Education	88%

Technical Skills

- **Languages:** Java, Python, C, TypeScript
- **CS Fundamentals:** Data Structures, Algorithms, OOP, System Design, Problem Solving
- **Backend & APIs:** REST APIs, Node.js, Express, FastAPI, Microservices
- **Databases:** SQL, MySQL, Relational Database Design
- **ML / AI:** Scikit-learn, TensorFlow, NumPy, Pandas, Model Training & Evaluation
- **Frontend:** HTML, CSS, JavaScript, React, Tailwind CSS
- **Tools & Practices:** Git, Linux

Projects

- | | |
|--|--|
| FinSight – AI-Powered Financial Intelligence Platform | <i>React, Node.js, Express, REST API</i> |
|--|--|
- Architected a **full-stack financial analytics platform** (React, Tailwind CSS, Node.js, Express) with an AI-driven OCR document processing pipeline, reducing manual analysis time and improving throughput by an estimated **40%**.
 - Designed a modular frontend with **20+ reusable UI components** following REST API and software development lifecycle best practices, enabling rapid and maintainable feature iteration.
- | | |
|--|---|
| DoseWise – AI-Powered Insulin Management System | <i>React, TypeScript, FastAPI, PostgreSQL, TensorFlow</i> |
|--|---|
- Built a scalable **full-stack healthcare platform** (React, TypeScript, FastAPI, PostgreSQL, TensorFlow) with ML models achieving **~89% prediction accuracy** for real-time glucose level forecasting and insulin dosing recommendations.
 - Integrated **Ethereum smart contracts** for secure, immutable medical record storage and designed a **relational database schema** in PostgreSQL ensuring high data integrity and application resiliency.
- | | |
|--|-----------------------------|
| AI-Based Fault Detection in Electrical Machines | <i>Python, Scikit-learn</i> |
|--|-----------------------------|
- Designed a **machine learning pipeline** for automated fault detection in electrical machines using Python and Scikit-learn, covering data preprocessing, feature engineering, and multi-class classification across **3+ fault categories**.
 - Conducted predictive maintenance analysis on real-world sensor datasets, improving fault detection reliability and reducing detection latency compared to manual inspection.

Experience

- | | |
|--------------------------------|---|
| Media and Creative Head | <i>IE(I) Student Chapter, VIT Chennai</i> |
|--------------------------------|---|
- Led media and design initiatives for **5+ technical workshops and events**, coordinating cross-functional teams to deliver digital content on schedule and strengthen the chapter's brand presence.
 - Managed promotional campaigns and social media engagement, demonstrating strong interpersonal, communication, and organizational skills in a collaborative environment.
- | | |
|---------------------|--|
| Board Member | <i>Mharo Rajasthan Club, VIT Chennai</i> |
|---------------------|--|
- Organized **multiple student events and cultural programs**, managing end-to-end logistics for audiences of **100+ students** and coordinating with diverse teams to drive community engagement.

Leadership & Achievements

- **Media and Creative Head** — IE(I) Student Chapter, VIT Chennai
- **School Prefect** — GD Birla School; selected among top students for demonstrated leadership, responsibility, and communication skills.